Spatiotemporal Association of Saffron Flowers For Bloom Prediction

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Project Goal

Build a system that tracks individual saffron flowers across multiple days to enable bloom prediction models.

Motivation

Alfalfa consumes 68% of Utah's water and produces 0.2% of the state's GDP.
Saffron has a water productivity ($ value / gal water) 8x greater than alfalfa. Replacing a portion of alfalfa with saffron could significantly reduce the total water consumption. However, no commercial equipment exists to harvest saffron, and high labor costs prevent saffron farming in the United States.
By creating models that map flower locations and predict maturity, we progress toward autonomous harvesting and enable the cultivation of this valuable crop.

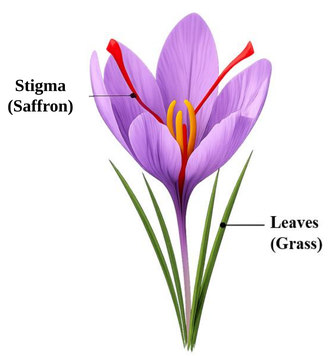

Figure 1. Anatomy of crocus sativus (saffron flowers)

Figure 2. Example of hand harvesting saffron

Data Collection

We planted a field of **15000** saffron plants in Fountain Green, Utah in September 2025. For data collection we outfitted a Farm_ng Amiga rover with 3 Intel RealSense D415 cameras, 2 Luxonis OAK-D W PoE cameras, Emlid Reach 2+ RTK-GPS, and a 5000 lumen LED array. The rover traversed the entire field 3 times daily for 23 days, capturing **1.6M** images.

Methods

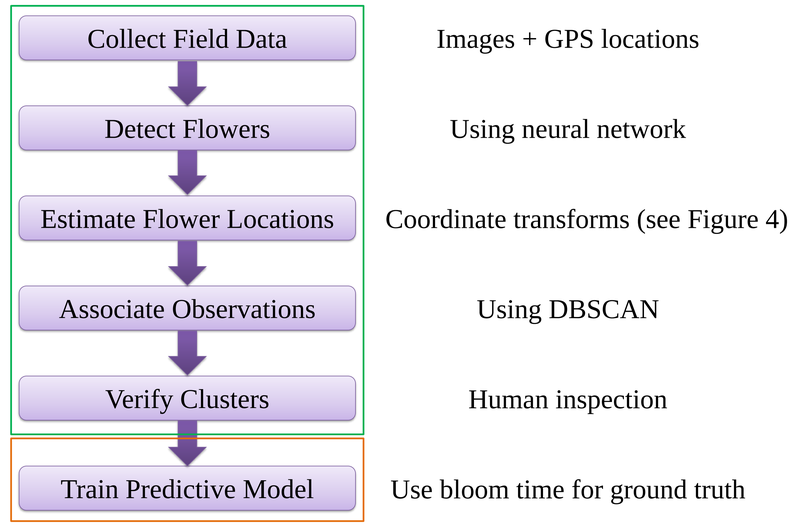

Figure 3. Process chart of the data pipeline. This work produces clustered images that will be used to predict bloom time.

After the images were collected, we trained a neural net to identify saffron flowers and report their location within the image. These pixel coordinates are projected into 3D space using a flat ground plane assumption. The geometric transform from the camera to the GPS antenna is known which allows a location estimate of every flower in each image. We used DBSCAN clustering algorithm to associate each measurement to its predicted landmark.

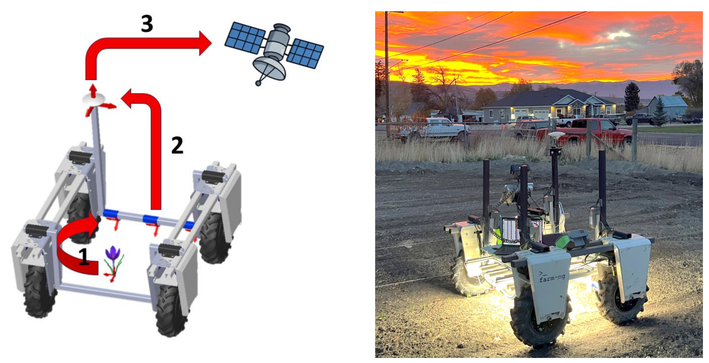

Figure 4. Left: Coordinate transforms: 1) 3D location to camera pixel. 2) Camera frame to antenna. 3) Antenna to global frame via RTK base station. Right: Data collection rover in the field at sunrise

Data and Results

From a test set of **48,000 flower detections**, the clustering process identified **1,490 plant landmarks**, compared to **1,516 actual flowers** present in the field during that time period, for a **counting error of 1.7%**.
To evaluate association quality, we randomly sampled **50 clusters** from high-density regions to manually verify. **96% of clusters** contained perfect sets with images of a single flower across multiple days.

Figure 5. Reconstructed flower growth from associated images.

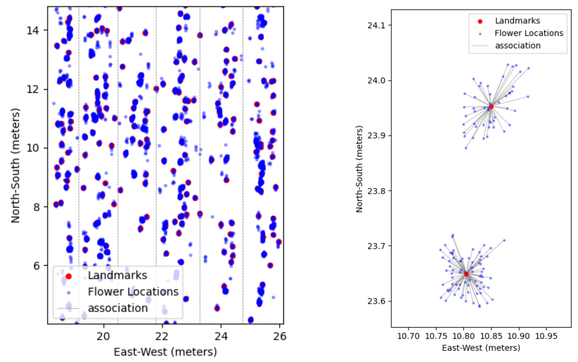

Figure 6. Left: Plot of flower measurements and their associated landmarks. Right: Zoomed in plots of individual clusters.

Conclusions

We developed a system that detects and associates individual saffron flowers across multiple days using only monocular cameras and RTK-GPS. Using geometric projection and clustering, flower observations were grouped with **96% cluster accuracy** in dense planting conditions.
This dataset enables the next stage of this work: training predictive models to estimate bloom timing and support future autonomous saffron harvesting systems.